

YatraLink — Product Requirements Document

Version 1.0 • MVP Product Definition

1. Product Overview

YatraLink is a railway journey-optimization platform for India. It helps passengers find practical ways to travel between two locations when a direct confirmed journey is unavailable. The core innovation is searching same-train split segments and connecting trains, then ranking valid journeys by availability, price, time, convenience and connection risk.

2. Problem Statement

Conventional railway search answers whether a train exists from A to B. When the full route has no confirmed availability, passengers often manually try intermediate stations, different trains and combinations. This is time-consuming and easy to miss good options.

3. Vision

Turn “no direct seat” into “here are the best practical ways to reach your destination.”

4. Goals

- Find direct trains and availability.
- Detect same-train split journeys.
- Find valid one-change and two-change journeys.
- Rank alternatives with explainable scoring.
- Show exactly which segments need separate bookings.
- Distinguish AVAILABLE, RAC, WAITLIST, NOT AVAILABLE and UNKNOWN.
- Eventually support authorized booking integration.

5. Non-Goals for MVP

- Replacing IRCTC.
- Unauthorized scraping, CAPTCHA bypass or personal-account automation.
- Guaranteeing confirmation or protecting separate-ticket connections.
- Native mobile apps, payments, PNR management and loyalty features.

6. Target Users

User	Need
Frequent traveler	Find reliable alternatives quickly.
Student	Find affordable confirmed combinations.

User	Need
Family traveler	Find practical journeys with fewer changes.
Business traveler	Optimize for time and convenience.

7. Primary User Flow

Enter origin → destination → date → passengers → class → search → direct availability check → connection engine if needed → rank options → show journey details → authorized booking pathway.

8. Search Inputs

Input	MVP
Origin / destination	Required
Journey date	Required
Passengers	Required; 1–6
Class	Optional/defaulted
Quota	Optional
Preference	Best overall / cheapest / fastest / fewest changes
Budget / departure window	Future enhancement

9. Core Features

9.1 Direct Search

Search direct trains and display schedule, class, fare and availability.

9.2 Same-Train Split

If BLR→DEL is unavailable, test intermediate stations on the same train. Separate reservations may be required even though there is no physical train change.

9.3 Connecting Trains

Search journeys with a train change. Validate arrival/departure timing and station compatibility.

9.4 Multi-Connection Search

Support up to two train changes in MVP.

9.5 Connection Risk

Classify connections using transfer time, station change and other deterministic risk factors.

9.6 Ranking

Rank valid journeys by availability, price, time, number of changes, same-train continuity and risk.

9.7 Explainability

Show segments, trains, transfer points, total time, total fare and availability.

10. Availability States

State	Meaning
AVAILABLE	Bookable inventory reported by the configured data source.
RAC	Reservation Against Cancellation.
WAITLIST	Waitlisted inventory.
NOT AVAILABLE	No inventory for requested segment/class/quota.
UNKNOWN	Data unavailable or stale; never treat as confirmed.

11. Recommendation Principles

Prefer feasible, low-risk journeys over theoretically cheaper but impractical journeys. Same-train split should normally outrank a physical train change when price and availability are comparable. Risky connections and station transfers receive penalties.

12. MVP Scope

- Responsive web app.
- Origin/destination/date/passenger/class search.
- Direct availability results.
- Same-train split detection.
- One- and two-change journey search.
- Fare, duration, transfer time and availability.
- Ranking and filters.
- Journey detail page.
- Graceful stale/unavailable data handling.
- Automated tests for the connection engine.

13. Success Metrics

Metric	Definition
Successful Journey Discovery Rate	Valid journey options found / difficult searches with no direct confirmed option.
Search completion rate	Searches resulting in viewed journey detail.
Recommendation selection rate	Searches where a user selects a recommended option.
Median search latency	Time from search submission to usable results.
Invalid-route rate	Generated routes rejected by validation.

14. MVP Definition of Done

- Search A→B works for selected date/class.
- Direct and segmented availability are represented accurately from the configured source.
- Same-train splits are detected.

- One/two-change routes are generated and validated.
- Impossible or unsafe connections are rejected or clearly flagged.
- Results are ranked and explainable.
- Unknown availability is never shown as confirmed.
- Core route-engine tests pass.
- Responsive UI works on desktop and mobile.